

How to Measure RSI

Nine data asks for frontier AI companies

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DATA TO PUBLISH



CAPABILITIES RSI would accelerate capability growth, and the acceleration would show up in internal models before released ones.

- Internal-model benchmark scores.** Results on a diverse set of benchmarks for the latest internally deployed models, including unreleased ones, aggregated into Epoch's Capabilities Index (ECI) or an internal equivalent. [Fig. 1](#)



EFFICIENCY Faster capability growth may only reflect more spending. Efficiency gains isolate better algorithms and data, which is where AI-assisted research would appear.

- Capability against training spend.** For each major model, capability as a function of total training expenditure, and the year-on-year efficiency gain at fixed capability. [Fig. 2](#)



- Pre-training compute efficiency.** Loss against pre-training compute for each production run, and for the small validation experiments that precede it. [Fig. 3](#)



- Post-training compute efficiency.** Capability against post-training compute, measured on ECI and on unsaturated AI R&D benchmarks. [Fig. 3](#)



INPUTS TO AI RESEARCH Gains driven by more human labor are not evidence of RSI; gains driven by more inference compute are. Each as a historical and ongoing series.

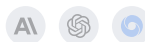
- Inference compute.** Usage in quantity (e.g. tokens) and in expenditure.



- Experimental compute.** Usage in quantity (e.g. H100 hours) and in expenditure.



- Headcount and compensation.** Employees and total compensation by team, function, and seniority.



- How AI generates efficiency gains.** Surveys of researcher uplift, and a record of autonomous discoveries.



- Research team budgets.** Budgets by team over time.



INTRODUCTION

Frontier AI companies have recently indicated willingness to share more information about recursive self-improvement (RSI). These are the nine disclosures that would tell us most, in priority order, drawing from [Cunningham et al. \(2026\)](#).

Figure 1. Hypothetical model capabilities over time. Internal models leave the historical trend first.

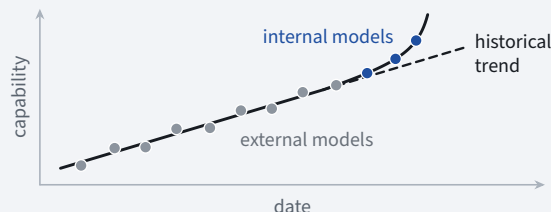
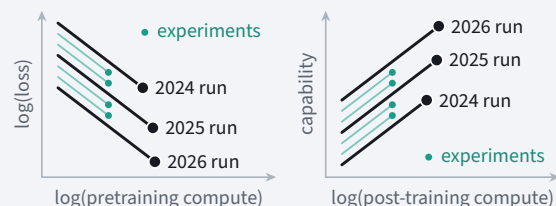


Figure 2. Hypothetical scaling curves of capability against training spend. Arrows show spend saved at equal capability.



Figure 3. Hypothetical pre-training loss (left) and post-training capabilities (right) against compute. Smaller experiments (teal) precede each production run.



SCORECARD



We score company disclosures against the nine asks: 1 for a full release, 0.5 for a partial release, 0 for no qualifying data found. Only 2026 measurements concerning models that were frontier at the time, or their development, qualify. Frontier includes leading general-purpose and advanced reasoning models; ask 1 requires unreleased models. See the companion scorecard for detailed explanations.

● released ○ partial ○ no credit